

Compact and efficient polarization rotator using laterally asymmetric rib waveguides on a lithium-niobate-on-insulator platform

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Photonic devices based on a lithium-niobate-on-insulator (LNOI) are current research hotspots; however, owing to the high refractive index contrast of the LNOI platform and inherent birefringence of lithium niobate itself, such photonic devices are generally polarization sensitive, affecting their further wide application. This paper proposes a simple, compact, and efficient polarization rotator (PR) based on a laterally asymmetric rib waveguide by depositing a layer of semi-infinite silicon nitride dielectric material on one side of the rib waveguide. The results show that a PR with a polarization rotation region length of $15.77\ \mu\text{m}$ is achieved, and the polarization extinction ratio (PER), insertion loss (IL), and polarization conversion efficiency (PCE) are 38.57/68.95 and 0.2/0.22 dB, and 99.99%/almost 100%, respectively, for the fundamental transverse electric mode (TE_0) and transverse magnetic mode (TM_0) at a $1.55\ \mu\text{m}$ wavelength. The operation bandwidth is around 120 nm for the TE_0 mode and TM_0 mode when the PER, IL, and PCE are greater than 20 dB, less than 0.32 dB, and more than 99%, respectively. Fabrication tolerances to the key structural parameters are investigated in detail. In addition, the evolution fields of the TE_0 mode and TM_0 mode along the propagation direction through the proposed device are presented. © 2023 Optica Publishing Group

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1. INTRODUCTION

Crystalline lithium niobate (LN) is an outstanding multifunctional material with excellent electro-optical properties, high second-order optical nonlinearity and acousto-optic properties, a wide transmission window from ultraviolet to mid-infrared, and stable physical and chemical characteristics, which can be said to be one of the most promising platforms for photonic integrated circuits (PICs) [1–4]. Recently, new developments and commercialization of lithium-niobate-on-insulator (LNOI) technology have opened a path for integrated on-chip photonic devices with unprecedented performance in terms of transmission loss, optical nonlinearity, and electro-optical tunability [5]. Successful manufactured wafer-level high-quality LN thin films on insulators and breakthroughs in nano-fabrication technology has made various on-chip high-performance structures [6–10], devices [11–16], and systems [1,17–19] a reality, and are revolutionizing traditional niobate lithium industry. The LNOI not only maintains the various excellent material properties of LN, but also provides strong optical limitations and a high integration degree of photonic devices due to its high refractive index. At present, the LNOI has successfully realized the hybrid integration with silicon (Si), silicon nitride (Si_3N_4) and other platforms [13,20–22]. It can

integrate the advantages of various materials and become a platform for the wide application of PICs in the future. However, photonic devices based on the LNOI are still in the stage of rapid development; to make the LNOI a more attractive and competitive PIC platform, more LNOI-based photonic devices need to be investigated. Similar to the polarization sensitivity problem based on the silicon-on-insulator (SOI) platform, LNOI-based photonic devices have also always suffered from polarization sensitivity due to the high refractive index contrast and inherent birefringence of the LN material itself. To overcome this problem, a polarization diversity scheme is often used [23]. In the polarization diversity system, the polarization rotator (PR) is one of the most fundamental devices, which is very important to realize effective polarization conversion. However, the research on polarization management devices based on the LNOI platform is still immature, so it is necessary and valuable to propose some devices with simple and compact structures while maintaining excellent performance.

In recent years, various kinds of schemes have been proposed to achieve polarization rotation, which can be mainly divided into two categories: mode-evolution-based and mode-interference-based. The former usually requires considerable device length to achieve high performance, while the latter

needs a relatively short device length [24–26]. Mode-evolution-based PRs [27,28] are not highly sensitive to the operating wavelength, which is beneficial for fabrication, but requires a considerable device length to achieve adiabatic evolution, which is not conducive to integration. Structures based on asymmetric directional couplers [29–32] are also widely adopted because they typically require only one-step etching, but they require different waveguide dimensions to break the coupler's symmetry, resulting in a large device area and complex manufacture processes. The PR based on mode interference not only can solve the compactness issue present in mode-evolution-based devices, but also can solve the volume and process fabrication problems from the asymmetric directional coupler structures [33]. Mode-interference-based PRs [24–26,33–36] can be smaller in size, and the key is to obtain asymmetric waveguides that can effectively rotate the optical axes. Currently, there are several methods to fabricate asymmetric waveguides, such as asymmetrically slanted-angle waveguides [37,38], asymmetrically etched waveguides [34,36], asymmetric slot-waveguides [24,25], and asymmetric hybrid plasmonic waveguides [26,33,35]. However, these methods result in a long device length or complicated manufacturing process for the LNOI platform, so their application is limited to a certain extent and is difficult to popularize. Recently, PRs based on the LNOI platform [33,39] have been proposed. The former is based on the principle of mode interference, but the device structure is relatively complex and, due to the inherent characteristic of hybrid plasmonic waveguides (homic loss caused by metal absorption), the insertion loss (IL) of the device is relatively large (1.44 dB). The latter is based on the principle of mode evolution, and the structure of the device is simple, but the length of the device is too long, which is very unfavorable for dense integration. Meanwhile, the performance of the device is not very good, and the polarization conversion efficiency (PCE) is only 95% and 99% when the device length is 100 and 150 μm , respectively.

In this paper, we propose a mode-interference-based LNOI PR that is compact, efficient, and easy to manufacture. By depositing a layer of semi-infinite Si_3N_4 dielectric material to one side of the LN rib waveguide, an asymmetric waveguide structure is introduced, resulting in the rotation of both orthogonal fundamental modes by a certain angle with respect to the optical axes. The two fundamental modes have different propagation constants and, after passing through the mode interference region, they are re-synthesized into another mode, which realizes the function of polarization rotation. The proposed device has a high polarization extinction ratio (PER) 38.57/68.95 dB, low IL 0.2/0.22 dB, and strong PCE 99.99%/almost 100% for the fundamental transverse electric mode (TE_0) and transverse magnetic mode (TM_0) at a 1.55 μm wavelength. The operation bandwidth is around 120 nm for both the TE_0 mode and TM_0 mode with the PER, IL, and PCE, respectively, being greater than 20 dB, less than 0.3 dB, and greater than 99%, while the length of the mode interference region is only 15.77 μm . In addition, we also discuss the influence of the operation wavelength and fabrication tolerances on device performance.

2. DEVICE STRUCTURE AND PRINCIPLE

A 3D view of our proposed PR is shown in Fig. 1(a), and the device contains three parts: input, output, and mode interference region (polarization rotation region). Figure 1(b) shows a cross-sectional view of the mode interference region. The input and output sections consist of the same type of LN rib waveguide, for which both sidewall angles are 70° . The width and height of the LN rib waveguide are w_r and h_r , respectively. In this paper, the w_r and h_r are set to 0.7 and 0.5 μm , respectively, which are commercially readily available. The middle mode interference region is composed of a LN rib waveguide with sidewall angles of $\alpha = 90^\circ$ and $\beta = 70^\circ$ and a semi-infinite Si_3N_4 dielectric material cover layer with a thickness of 0.152 μm , respectively. By introducing a Si_3N_4 cover layer, the symmetry of the waveguide is significantly broken and by optimizing the thickness of Si_3N_4 cover layer, an optimal optical axis rotation angle of 45° with respect to the horizontal and vertical directions can be obtained to get a high PCE. As shown in Fig. 1(a), we selected an X -cut/ Y -propagation LNOI wafer, whose extraordinary refractive index (n_e) is 2.138 along the Z axis, and ordinary refractive index (n_o) is 2.211 along the X and Y axis at a 1.55 μm wavelength. Furthermore, the refractive indices of silicon dioxide (SiO_2) and Si_3N_4 are 1.444 and 1.996 [40] at the wavelength of 1550 nm, respectively.

To study the two orthogonal fundamental mode characteristics in the mode interference region and choose the appropriate structural parameters, a full-vectorial mode solver based on the finite-difference frequency-domain (FDFD) method is utilized [41]. The semi-infinite Si_3N_4 dielectric material cover layer breaks the symmetry of the waveguide structure, resulting in two orthogonal fundamental modes, Mode1 and Mode2; the magnetic field profiles of the two fundamental modes are shown in Figs. 2(a) and 2(b), and the white arrow is the direction of the magnetic field. Figures 2(c) and 2(d) illustrate the two magnetic field components H_y and H_z of Mode1, wherein the direction of H_y is the positive direction of the y axis, and the direction of H_z is the negative direction of the z axis. Therefore, the magnetic field direction of Mode1 synthesized by H_y and H_z is rotated about 45° relative to the original optical axes. Similarly, Figs. 2(e) and 2(f) illustrate the two magnetic field components H_y and H_z of Mode2, wherein the direction of H_y is the positive direction of the y axis, and the direction of H_z is the positive direction of the z axis as well. Consequently, the magnetic field direction of Mode2 synthesized by H_y and H_z is rotated about

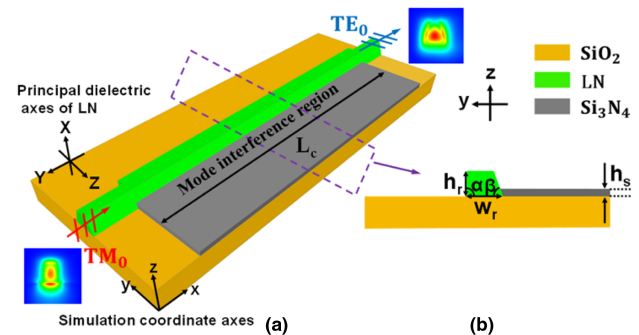


Fig. 1. Schematic of the PR. (a) 3D view and (b) cross-sectional view of the polarization rotation region. The substrate of Si or LN and upper cladding of air are not shown here for clarity.

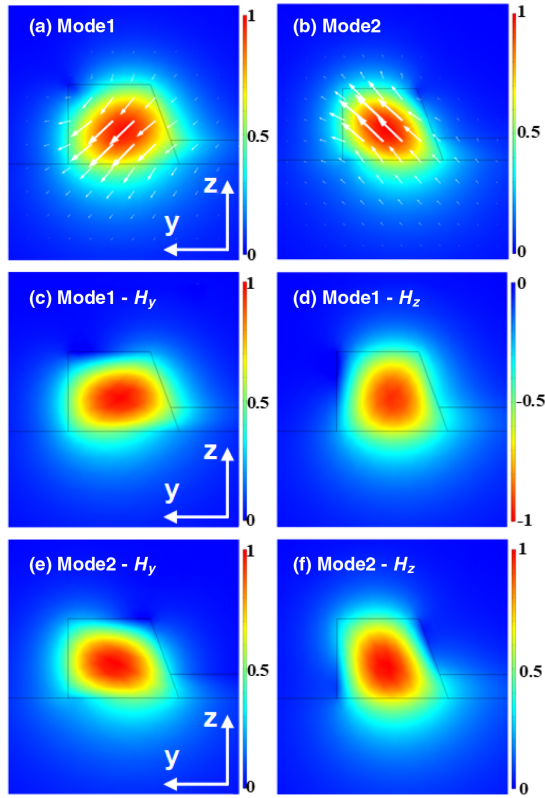


Fig. 2. Magnetic field and magnetic field component profiles of the two orthogonal fundamental modes. (a) Mode1 and (b) Mode2. The white arrow is the direction of the magnetic field. Magnetic field components (c) H_y and (d) H_z for Mode1. Magnetic field components (e) H_y and (f) H_z for Mode2. Here $w_r = 0.7 \mu\text{m}$, $h_r = 0.5 \mu\text{m}$, $h_s = 0.152 \mu\text{m}$, $\alpha = 90^\circ$, and $\beta = 70^\circ$.

45° relative to the original optical axes. Compared to the SOI platform [25,26], the LNOI has a lower refractive index, resulting in a larger mode area and a longer device length. To reduce the size of the PR, we decided to use LN with a slanted-angle sidewall [37]. It produces a stronger optical axis rotation compared to the LNOI PR in which both sidewall angles 90° . For example, if the TM_0 polarization light is used as the input mode, two orthogonal fundamental modes will be excited in the mode interference region, and the two fundamental modes will beat each other during the transmission process. Due to the different propagation constants of the two fundamental modes, the phase difference is constantly changing during the propagation process; when the phase difference reaches π , it is equivalent to one of the fundamental modes flipped by 180° and, after passing through the mode interference region, the two fundamental modes are recoupled into a new mode, which is rotated by 90° compared to the input mode; that is, the input TM_0 mode has changed to the TE_0 mode. By reasonably designing and optimizing the thickness of the semi-infinite Si_3N_4 cover layer, the optical axes of the two fundamental modes can be rotated by 45° relative to the y axis and z axis, and the length of the mode rotation region $L_c = L_\pi$ (L_π is half-beat length) is calculated as follows [37]:

$$L_c = \frac{\pi}{\beta_1 - \beta_2} = \frac{\pi}{k_0(n_{\text{eff}1} - n_{\text{eff}2})}, \quad (1)$$

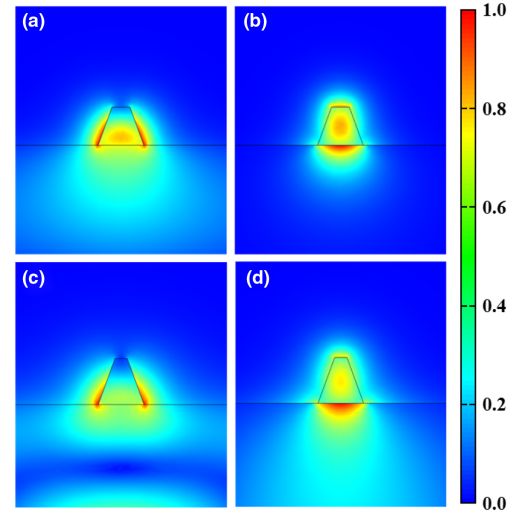


Fig. 3. Electrical field profiles for the TE_0 mode and TM_0 mode. (a), (b) When $w_r = 0.6 \mu\text{m}$ and (c), (d) when $w_r = 0.5 \mu\text{m}$.

where L_c is the half-interference length, β_1 and β_2 are the propagation constants of the two fundamental modes, k_0 is the vacuum wave number, and $n_{\text{eff}1}$ and $n_{\text{eff}2}$ are the effective refractive indices of the two fundamental modes. Of course, this structure also applies to the TE_0 input mode.

When the height of the LN rib waveguide, h_r is fixed at $0.5 \mu\text{m}$, and the sidewall angles α and β are equal to 70° . On one hand, as the width w_r decreases (less than $0.6 \mu\text{m}$), as shown in Fig. 3, we find that the light power can no longer be efficiently limited in the LN rib waveguide, and the transmission loss is large. On the other hand, as the width w_r increases (greater than $0.8 \mu\text{m}$), the tuning effect of the semi-infinite Si_3N_4 cover layer becomes weaker and weaker. When $w_r = 0.8 \mu\text{m}$, the half-interference length has reached $56.2 \mu\text{m}$ and, when $w_r = 0.9 \mu\text{m}$, the rotation angle of the two orthogonal fundamental modes cannot reach 45° , no matter how we adjust the thickness of the semi-infinite Si_3N_4 cover layer. In addition, considering the single-mode transmission condition, we determined the width of the LN rib waveguide to be $0.7 \mu\text{m}$. For the specified w_r ($0.7 \mu\text{m}$ in this article), in the input and output parts of the proposed PR, when the height of the rib waveguide is less than $0.4 \mu\text{m}$, the TM_0 mode is gradually cut off. When h_r takes 0.5 , 0.6 and $0.7 \mu\text{m}$, to rotate the two fundamental modes by 45° relative to the optical axes, the thicknesses of the corresponding Si_3N_4 cover layer are 0.175 , 0.27 , and $0.332 \mu\text{m}$, and the lengths of the corresponding mode interference region are 28.85 , 29.39 , and $38.61 \mu\text{m}$, respectively, according to Eq. (1). As a result, it was determined that the height h_r was $0.5 \mu\text{m}$. When $w_r = 0.7 \mu\text{m}$, $h_r = 0.5 \mu\text{m}$, and $\beta = 70^\circ$, we obtain the required semi-infinite Si_3N_4 cover layer's thickness for the two orthogonal fundamental modes to be rotated by 45° with respect to the optical axes, effective refractive indices of the two fundamental modes, and half-interference length at different angles of α . As shown in Fig. 4(a), with the increase of α , the effective refractive index difference of the two fundamental modes becomes larger and larger so, according to Formula (1), we know that the half-interference length L_c is getting smaller and smaller, and the result is shown in Fig. 4(b). Therefore, we know that, with the increase of the lateral asymmetry of

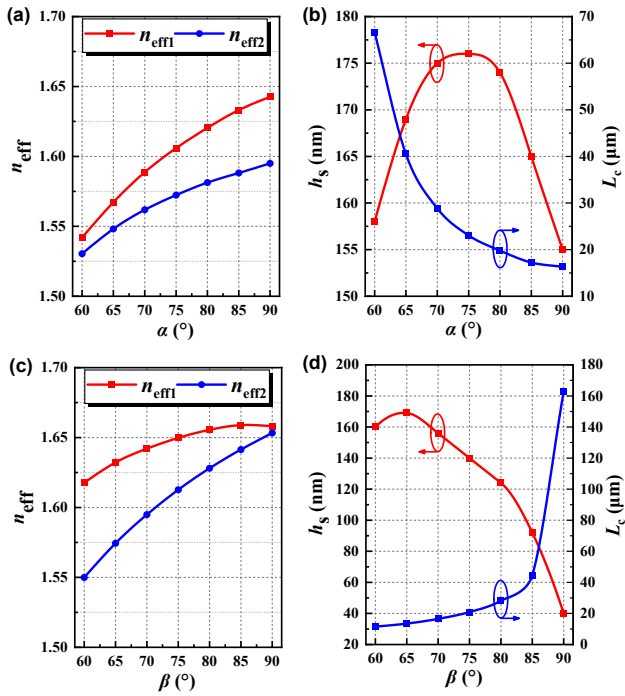


Fig. 4. Effective refractive indices of the two orthogonal fundamental modes; required Si_3N_4 cover layer thickness and half-interference length as functions of α and β . (a), (b) When $\beta = 70^\circ$, α as a variable; (c), (d) when $\alpha = 90^\circ$, β as a variable. Here $w_r = 0.7 \mu\text{m}$, and $h_r = 0.5 \mu\text{m}$.

the LN rib waveguide in the polarization rotation region, the half-interference length becomes smaller and smaller. On the other hand, we also tested the cases where α takes 90° , and β takes different angles; the results are shown in Figs. 4(c) and 4(d). It should be noted that, in order to maintain w_r constant while changing β , the top width of the waveguide needs to be changed accordingly. We see that with the increase of β , the lateral asymmetry of the LN rib waveguide in the polarization rotation region gradually decreases, and the half-interference length becomes larger and larger. However, when β is 60° and 65° , although the length of the polarization rotation region is short, the transmission loss of the TE mode is relatively large. As a result, we choose $\beta = 70^\circ$ as a compromise. According to the above analysis, we finally determined the parameters of the device as $w_r = 0.7 \mu\text{m}$, $h_r = 0.5 \mu\text{m}$, $\alpha = 90^\circ$, and $\beta = 70^\circ$.

To evaluate the performance of the designed PR, the PCE is an important parameter, which is defined as the ratio of the desired polarization to the sum of the two polarizations at the output port:

$$\text{PCE}_{(\text{TM} \rightarrow \text{TE})} = \frac{P_{\text{TE}}}{P_{\text{TE}} + P_{\text{TM}}} * 100\%, \quad (2)$$

$$\text{PCE}_{(\text{TE} \rightarrow \text{TM})} = \frac{P_{\text{TM}}}{P_{\text{TE}} + P_{\text{TM}}} * 100\%. \quad (3)$$

In addition, for mode-interference-based PRs, the PCE is related to the optical axis rotation angle of the two orthogonal fundamental modes [37]:

$$\text{PCE} = \sin^2(2\theta) \sin^2\left(\frac{\pi L}{2L_c}\right) * 100\%, \quad (4)$$

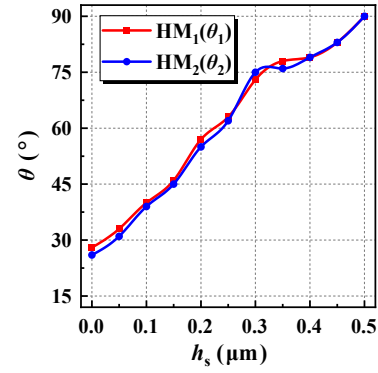


Fig. 5. Two orthogonal fundamental mode optical axis rotation angles as a function of the thickness of the Si_3N_4 cover layer. Here $w_r = 0.7 \mu\text{m}$, $h_r = 0.5 \mu\text{m}$, $\alpha = 70^\circ$, and $\beta = 90^\circ$.

where L is the mode conversion length, and θ is the optical axis rotation angle of the two fundamental modes, defined as [37]

$$\tan(\theta) = \frac{\iint \varepsilon(y, z) E_y^2(y, z) dy dz}{\iint \varepsilon(y, z) E_z^2(y, z) dy dz}, \quad (5)$$

where $\varepsilon(y, z)$ is the real part of the dielectric constant distribution, and $E_y(y, z)$ and $E_z(y, z)$ are the electric field components in the horizontal and vertical directions of the two fundamental modes, respectively. Note that, because of the anisotropy of the LN material itself, $\varepsilon(y, z)$ should be replaced with $\varepsilon_y(y, z)$ and $\varepsilon_z(y, z)$. According to Eq. (4), it is obvious that $\theta = 45^\circ$ is necessary to obtain a PCE of 100%. For mode-interference-based PRs, it is important to maintain the balance of the two fundamental modes, which directly affects the performance of the device. The rotation angle θ of the two fundamental modes affects the PCE and the balance of the two HMs, while the rotation angle θ is dependent on w_r , h_r , and h_s . Figure 5 shows the variation characteristics of the optical axis rotation angles θ_1 and θ_2 with the thickness of the Si_3N_4 cover layer h_s , when the width w_r and the height h_r of the LNOI rib waveguide is 0.7 and $0.5 \mu\text{m}$, where θ_1 and θ_2 are the rotation angles of Mode1 and Mode2, respectively. From the figure, we see that θ_1 and θ_2 experience a growth change from approximately 30° to 90° as the thickness of the Si_3N_4 cover layer increases. To achieve the balance of the two orthogonal fundamental modes, $\theta_1 = \theta_2$ is required. Unfortunately, for the specified w_r , h_r , and h_s , $\theta_1 = \theta_2 = 45^\circ$ is not always satisfied. However, by calculating the PCE according to Formula (4), we find that—when $|\theta - 45^\circ| \leq 2.85^\circ$, the maximum PCE can exceed 99%, which is obviously acceptable.

3. RESULTS AND DISCUSSION

In order to verify the design principle and demonstrate the expected performance of the present device, the optical transmission in the proposed PR is calculated numerically using the three-dimensional finite difference time domain (3D-FDTD) method. Here we set $w_r = 0.7 \mu\text{m}$, $h_r = 0.5 \mu\text{m}$, $h_s = 0.155 \mu\text{m}$, $\alpha = 90^\circ$, and $\beta = 70^\circ$. When the input source power is normalized to 1 (0 dB), the PER and IL are defined as

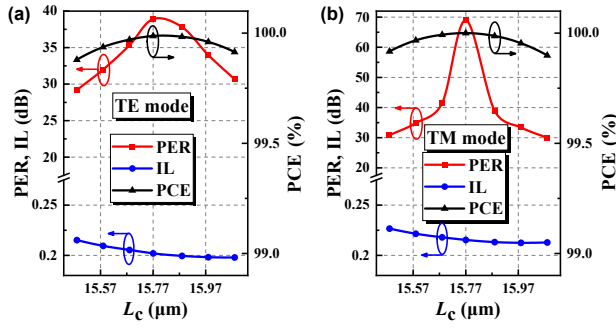


Fig. 6. PER, IL, and PCE of the proposed PR as a function of L_c at the wavelength of $1.55 \mu\text{m}$ for the (a) TE mode and (b) TM mode. Here $w_r = 0.7 \mu\text{m}$, $h_r = 0.5 \mu\text{m}$, $h_s = 0.152 \mu\text{m}$, $\alpha = 90^\circ$, and $\beta = 70^\circ$.

$$\text{PER}_{(\text{TM} \rightarrow \text{TE})} = 10 \log_{10} \left(\frac{P_{\text{TE}}}{P_{\text{TM}}} \right), \quad (6)$$

$$\text{PER}_{(\text{TE} \rightarrow \text{TM})} = 10 \log_{10} \left(\frac{P_{\text{TM}}}{P_{\text{TE}}} \right), \quad (7)$$

$$\text{IL}_{(\text{TM} \rightarrow \text{TE})} = -10 \log_{10} \left(\frac{P_{\text{TE}}}{P_{\text{input}}} \right), \quad (8)$$

$$\text{IL}_{(\text{TE} \rightarrow \text{TM})} = -10 \log_{10} \left(\frac{P_{\text{TM}}}{P_{\text{input}}} \right), \quad (9)$$

where P_{TE} , P_{TM} are the output power of TE mode and TM mode at the output port, and P_{input} is the input power of the inject mode (TE mode or TM mode) in the input port.

As shown in Figs. 6(a) and 6(b), we know that the optimal length of the polarization rotation region by 3D-FDTD simulation is $15.77 \mu\text{m}$ for both TE_0 and TM_0 input modes, which is in good agreement with the result obtained in Formula (1). To demonstrate the transmission features of the proposed PR, the evolution profiles of the electric field dominant components of the injected TE_0 mode (y -polarized) and TM_0 mode (z -polarized) are illustrated in Fig. 7 with the parameters of $w_r = 0.7 \mu\text{m}$, $h_r = 0.5 \mu\text{m}$, $h_s = 0.152 \mu\text{m}$, $\alpha = 90^\circ$, $\beta = 70^\circ$, and $L_c = 15.77 \mu\text{m}$ in the polarization rotation region. It can be seen clearly that when the TE_0/TM_0 mode is launched, the electric field principle component E_y/E_z gradually disappears in the mode interference region, and E_z/E_y begins to appear slowly and gradually becomes dominant in the output region. Consequently, we realized the function of polarization rotation.

Figure 8 shows the influence of the wavelength on device performance. For the TE_0 mode incident, at the wavelength of 1557 nm , the calculated PER and IL are 55.83 and 0.21 dB , respectively. The bandwidth is about 40 nm for a $\text{PER} > 30 \text{ dB}$, $\text{IL} < 0.23 \text{ dB}$, and the bandwidth is about 120 nm for a $\text{PER} > 20 \text{ dB}$, $\text{IL} < 0.32 \text{ dB}$. Similarly, for the TM_0 mode incident, at the wavelength of 1550 nm , the calculated PER and IL are 68.95 and 0.22 dB , respectively. The bandwidth is about 40 nm for a $\text{PER} > 30 \text{ dB}$, $\text{IL} < 0.23 \text{ dB}$, and the bandwidth is about 120 nm for a $\text{PER} > 20 \text{ dB}$, $\text{IL} < 0.31 \text{ dB}$.

In addition, we have analyzed the impact of fabrication tolerances on device performance. Figure 9 shows the PER, IL, and

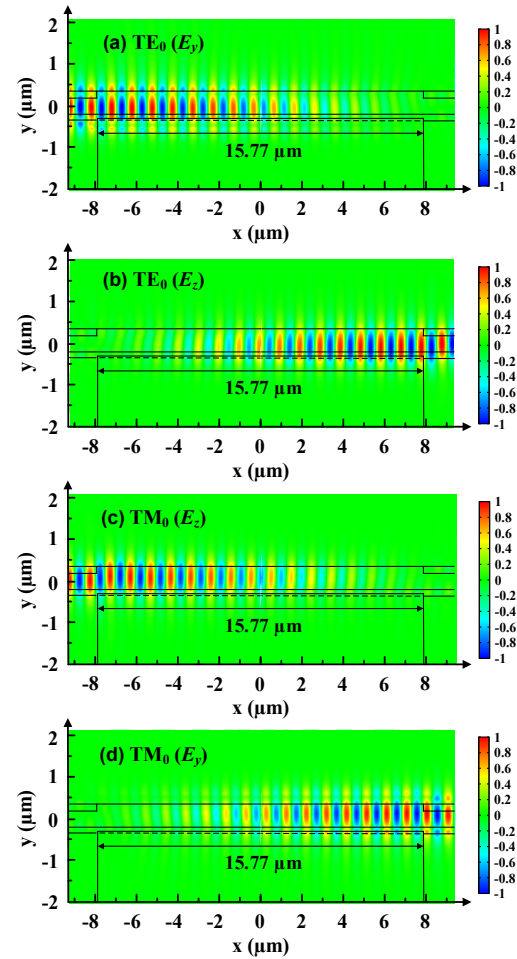


Fig. 7. Electric field evolution profiles of the designed PR at a $1.55 \mu\text{m}$ wavelength for (a) E_y , (b) E_z , (c) E_z , and (d) E_y when the TE_0 mode and TM_0 mode are input from the left port.

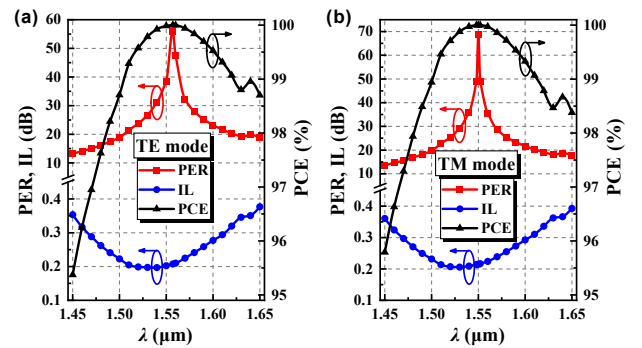


Fig. 8. Wavelength dependence of the device performances (PER, IL, and PCE) for the (a) TE mode (b) TM mode.

PCE as functions of the sidewall angle variations $\Delta\alpha$ and $\Delta\beta$ of the LN rib waveguide in the polarization rotation region at the wavelength of $1.55 \mu\text{m}$. It can be seen that the PR exhibits a large enough tolerance to the sidewall angle α (approximately $-2.8^\circ < \Delta\alpha < 0^\circ$) to make a PER greater 20 dB , IL less than 0.23 dB , and PCE greater than 99% for both input modes. Similarly, the device exhibits a large tolerance to the sidewall angle β (approximately $-1.5^\circ < \Delta\beta < 1.5^\circ$) to make a PER

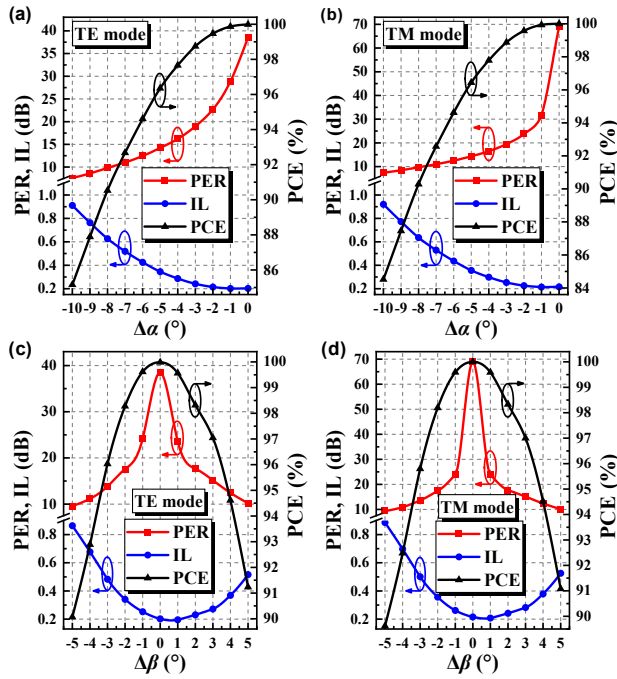


Fig. 9. PER, IL, and PCE of the proposed PR as functions of fabrication tolerances $\Delta\alpha$, $\Delta\beta$ for the (a), (c) TE mode and (b), (d) TM mode.

greater 20 dB, IL less than 0.23 dB, and PCE greater than 99% for both input modes. Figure 10 shows the effects of the thickness error Δh_s of the Si₃N₄ cover layer and the length (that is, the length of the semi-infinite Si₃N₄ cover layer) error ΔL_c of the polarization rotation region on the PER, IL, and PCE at the wavelength of 1.55 μm , respectively. From Fig. 10, we see that the device has a high tolerance for the semi-infinite Si₃N₄ cover layer. As can be seen from Figs. 10(a) and 10(b), the PR exhibits a large tolerance to the thickness of the Si₃N₄ cover layer (approximately $-15 \text{ nm} < \Delta h_s < 12 \text{ nm}$) to make a PER greater than 20 dB, IL less than 0.25 dB, and PCE greater than 99% for both input modes. As can be seen from Figs. 10(c) and 10(d), the PR exhibits a large tolerance to the length of the Si₃N₄ cover layer (approximately $-1 \mu\text{m} < \Delta L_c < 1 \mu\text{m}$) to make a PER greater than 20 dB, IL less than 0.25 dB, and PCE greater than 99% for both input modes. Consequently, the designed PR is easy to manufacture and has stable performance. Furthermore, the PR we designed requires only one step of etching, and there will be no accumulative error.

Finally, it is essential to do a comparison of a presently proposed PR with other reported LN-based PRs. Table 1 brings together their structures, mode conversion type, principle, PER, IL, and device length. Reference [33] realizes the conversion from TM₀ to TE₀ based on the principle of mode interference. The device length is short, and the PER is high, but the IL is large due to the use of the hybrid plasmonic waveguide structure. Reference [39] realizes the conversion from TM₀ to TE₀ based on the principle of mode evolution; the device length is too long to be highly integrated, and the performance of the device is not very good. Reference [42] is based on the principle of the electro-optic effect and requires an electrode structure; the length of the device is as high as 2.5 mm, which cannot be

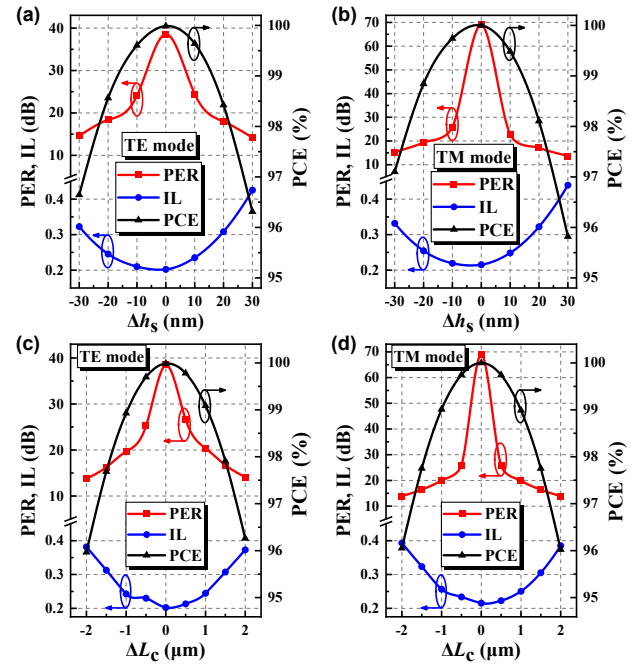


Fig. 10. PER, IL, and PCE of the proposed PR as functions of fabrication tolerances Δh_s , ΔL_c for (a), (c) TE mode and (b), (d) TM mode.

highly integrated. Reference [43] is based on a coupled-mode theory; the length of the device is as high as 1 mm and cannot be easily integrated. Reference [44] realizes the PR based on the principle of mode interference; the device length is short, the PCE is high, and the IL is small. However, in our work, the mutual conversion of TM₀ and TE₀ is achieved by adopting the principle of mode interference, and the device has a short length and excellent performance.

4. CONCLUSION

In this paper, based on the LNOI platform, a semi-infinite Si₃N₄ dielectric material cover layer is introduced on one side of the rib waveguide, and we successfully demonstrate a mode-interference-based PR. The structure has been investigated in detail to realize the function of polarization rotation with high performance by using full-vectorial FDFD and 3D-FDTD methods. The device has the advantages of a simple, compact structure and an easy fabrication process while owning excellent performance. From the numerical results, a polarization rotation region length of 15.77 μm with the PER, IL, and PCE being 38.57/68.95 and 0.2/0.22 dB and 99.99%/almost 100%, respectively, for fundamental TE₀ and TM₀. The designed PR shows a broadband performance (about 120 nm) to make a PER greater than 20 dB, IL less than 0.3 dB, and PCE greater than 99% for both the TE₀ mode and TM₀ mode. Furthermore, field evolution through the PR has also been presented for the both fundamental modes. Finally, the proposed PR is devised based on a conventional LNOI platform, which exhibits a great application potential as a building block in ultra-compact on-chip PICs.

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Table 1. Comparison of the Proposed PR with other Reported LN-Based PRs^a

Structures	Mode Conversion Type (principle)	PER(dB)/PCE(%)	IL (dB)	Device Length (μm)	Sources
Asymmetric rib HPW	TM ₀ → TE ₀ (Mode interference)	59.58	1.44	13.7	[33]
PW with semi-infinitecover layer	TM ₀ → TE ₀ (Mode evolution)	19.96	—	150	[39]
Ti-diffused waveguide	TE ₀ → TM ₀ TM ₀ → TE ₀ (Electro-optic effect)	23.6	—	25000	[42]
Tilted section with two straight segment waveguides	TE ₀ → TM ₀ TM ₀ → TE ₀ (Mode coupling)	—	—	1000	[43]
Asymmetric rib waveguide with semi-infinitecover layer	TE ₀ → TM ₀ TM ₀ → TE ₀ (Mode interference)	99.6% 99.2%	0.38 0.4	17.7	[44]
Asymmetric rib waveguide with semi-infinitecover layer	TE ₀ → TM ₀ TM ₀ → TE ₀ (Mode interference)	38.57 68.95	0.2 0.22	15.77	This work

^aHPW: hybrid plasmonic waveguides.

*PW: photonic wire.

**“—”: not mentioned.

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